

Instruction: 1) The ProgMods tab can be copied (as required) to create multiple ProgMods tabs, according to the programming module in the study.
2) Update the "Title" of the ProgMods to be descriptive (e.g., 5A.Baseline Characteristics)
3) List the newly added ProgMods tabs in Section C of the Cover tab
4) Delete any examples in grey italic text that are not relevant/applicable to the study once the programming specification has been finalized

Section A: Defining Medication Available Period (MAP) medications						
Title	10.LOT1_BASE_END					
Time Period	Variable	Label	Values	Definition	Code Lists Group	Additional Notes
	LOT1_END_DT_TEMP	Temporary end date for LOT1 base period		The LOT will continue until the earliest of any of the following: (1) Discontinuation of all agents in the regimen (with or without switch to a new agent); the LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available; (2) SCT / CAR-T events: if an autologous SCT occurs > 180 days following a previous autologous SCT, the LOT will end the day before the latter SCT. Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before the allogeneic SCT. CAR-T cellular therapy infusions are classified as their own LOT; the preceding LOT transition for CAR-T is defined in the LOT1_TX_ENDDATE_REASON and LOT1_BASE_END_DT rows. (3) Death; (4) Health plan disenrollment; (5) End of the study period. Note: Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT (this is an exception, not an ending event). The LOT will continue until the earliest of any of the following: (1) Discontinuation of all agents in the regimen, with or without switch to a new agent. The LOT end date is the run-out date, defined as the last date that any MM medication in that LOT is considered to be available, if the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration / dispense date of the new agent. (2) SCT / CAR-T events: if an autologous SCT occurs > 180 days following a previous autologous SCT, the LOT will end the day before the latter SCT. Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before the allogeneic SCT. CAR-T cellular therapy infusions are classified as their own LOT; the preceding LOT transition for CAR-T is defined in the LOT1_TX_ENDDATE_REASON and LOT1_BASE_END_DT rows. (3) Death; (4) Health plan disenrollment; (5) End of the study period. Note: Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT (this is an exception, not an ending event). Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. SCT is operationalized as >1 procedure code for an SCT. A single autologous SCT is one that is not part of a valid tandem pair. Single autologous SCTs and tandem SCTs are not unplanned SCTs and are considered a continuation of the line of therapy.		LOT1_BASE_END_DT derived from LOT1_BASE_DISCON_DT (max MAP_END_DT from T_MEDICAL_PROC_CD/NDC and T_RX_NDC for induction meds), LOT1_BASE_1ST_ADD_MED_DT (first MAP_START_DT from T_MEDICAL_TX_RX for non-induction meds minus 1), LOT1_TX_ENDDATE (earliest LOT-ending SCT from T_MEDICAL procedure codes minus 1), and OBS_END_DT (LEAST of study_end from config, DEATH_DT from T_DOD_VNDDO, disenrollment from T_MEMBER_CONF_ENROLLMENT_ELUGEND).
RU_PD	LOT1_END_REASON_TEMP	Temporary LOT1 base period end reason				End reason derived from matching the LOT1 end date against dates from: T_MEDICAL_TX_RX (discontinuation via MAP_END_DT), T_MEDICAL_TX (new med via MAP_START_DT), T_DOD (death via VNDDO), T_MEDICAL (SCT via procedure codes in T_MED_PROCEDURE), ENDDATE from T_MEMBER_CONF_ENROLLMENT (disenrollment) and study_end config.
RU_PD						JM 15Apr 2026
RU_PD						JM 06Apr 2026
RU_PD	LOT1_HSCT_AUTO_SIMG_FL	Flag: base period had single valid autologous HSCT				Autologous HSCT identified from T_MEDICAL procedure codes (PROC_CD_BILL_PROC_CD) matching SCT code list entries for AUTO type. Claims grouped by 14-day windows from T_MEDICAL_FST_DT. Gap validation (60 days) and tandem detection (180 days) applied to grouped dates.
RU_PD	LOT1_HSCT_AUTO_TAND_FL	Flag: base period had tandem valid autologous HSCT				Tandem detection uses T_MEDICAL procedure codes for both AUTO events. The 180-day window is applied to grouped AUTO dates (after 14-day windowing and 60-day gap validation). ALLO events between AUTO_DT_1 and AUTO_DT_2 checked via T_MEDICAL.
RU_PD	LOT1_TX_AUTO_MAX_DT	Latest date of patient's valid HSCT				Derived from T_MEDICAL procedure code dates for autologous HSCT, after 14-day windowing and 60-day gap validation.
RU_PD						JM 15Apr 2026

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FU_PD	LOT1_TX_AUTO_FLAG	Flag: patient had any valid autologous HSCT		Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. This binary variable indicates whether the patient received any valid autologous transplant, regardless of single or tandem.			Derived flag from T_MEDICAL procedure code analysis for autologous HSCT events.		JM 06Apr/2026
	LOT1_BASE_MEOS	LOT1 induction medications		The LOT1 regimen is defined as all multiple myeloma (MM) oncology therapies received on and within 90 days of the LOT1 start date, with the first therapy received considered the induction therapy. This variable therefore captures the full set of oncology agents administered during this 90-day window, excluding non-oncology agents such as corticosteroids. Switches between a biologic reference product and its biosimilars(s), or between biosimilars of the same reference product, are permitted and do not constitute a change in regimen or advance the line of therapy (Madduri, 2021).			Medications identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MM4_CODELIST (Tab 41). Medication abbreviations from CL_MM4_ROLLUP (Tab 40). Claims filtered by MAP_START_DT within induction window from LOT1_START_DT.		JM 06Apr/2026
FU_PD	LOT1_BASE_END_DT	Final LOT1 base period end date		LOT1_BASE_END_DT is the earliest applicable LOT-ending date derived from: - discontinuation of all agents, new qualifying medication addition, - qualifying SCT or CAR-T events, - death - health plan disenrollment - end of study period.			LOT1_BASE_END_DT derived from priority evaluation of dates from T_MEDICAL/ RX (MAP_END_DT for discontinuation, MAP_START_DT for new med), T_MEDICAL procedure codes (SCT events), T_DOD (YMDOD for death), T_MEMBER_CONT_ENROLLMENT (ELGEND for disenrollment), and study_end config parameter.		
FU_PD				Permissible substitutions do NOT end the LOT. Maintenance is NOT an independent LOT-ending event in this study. If the current LOT is immediately interrupted by a new qualifying medication, the LOT end date is the day before the first administration or dispense date of that new medication. For CAR-T transitions – including the CAR_T_INIT case – LOT1_BASE_END_DT is the day before FIRST_CART_DT. The CAR-T LOT begins on FIRST_CART_DT.					JM 06Apr/2026
	LOT1_BASE_END_REASON	Final LOT1 base period end reason		Final LOT1 base period end reason. Derived by selecting the earliest end date across the LOT-ending events listed in the LOT1_END_REASON_TEMP rules narrative			End reason derived from evaluation of dates across T_MEDICAL/ RX (medication events), T_MEDICAL (SCT procedure codes), T_DOD (death), and T_MEMBER_CONT_ENROLLMENT (disenrollment). Priority logic applied analytically.		JM 06Apr/2026
FU_PD				Priority/order below applies when more than one end-reason resolves on the same earliest end date: SCT_ALLO > SCT_CART > SCT_AUTO > CAR_T_INIT > MED_ADD > DISCONTINUATION > DEATH > DISENROLLMENT > STUDY_END					
				CART_INIT applies when a CAR-T infusion (FIRST_CART_DT) occurs within 45 days of the start date of the first added agent. The end reason is CART_INIT (not MED_ADD), and LOT1_BASE_END_DT is the day before FIRST_CART_DT.					JM 06Apr/2026
				This study does not use any Maintenance as end values					
FU_PD	FIRST_ALLO_DT	Date of first allogeneic HSCT during LOT1		Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. Rule 2 (SCT/CART events): Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.			ALLO SCT identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=ALLO. Date from FST_DT in T_MEDICAL. Sequential ordering applied to identify first ALLO event.		JM 06Apr/2026
FU_PD	FIRST_CART_DT	Date of first CAR-T therapy during LOT1		CAR-T: CAR-T cellular therapy infusions are classified as their own LOT. Therapies (including supportive agents, e.g., corticosteroids) given within 45 days of CAR-T cellular therapy are consolidated as part of the CAR-T LOT.			CAR-T identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=CAR-T. Date from FST_DT in T_MEDICAL.		JM 15Apr/2026
FU_PD	LOT_DISCON_GAP_DAYS	Discontinuation gap threshold (days)		Discontinuation of the LOT1 regimen occurs when all MM oncology agents in the line of therapy are stopped. If the regimen is discontinued and no new agent is initiated, the LOT end date is defined as the run-out date of the last administered agent. Sensitivity analyses will assess the robustness of the line of therapy rules by varying key algorithm parameters			Gap calculated as datediff(OBS_END_DT, RAW_DISCON_DT) where OBS_END_DT is derived from T_MEMBER_CONT_ENROLLMENT (ELGEND), T_DOD (YMDOD), and study_end config. RAW_DISCON_DT is max(MAP_END_DT) from T_MEDICAL/ RX for induction medications.		JM 15Apr/2026
FU_PD	MEDICAL_DAY_SUPPLY	Assumed days supply for medical claims		The days supply on medical claims is always missing and therefore a 28-day supply is assumed. When pharmacy claims are missing days supply, or when days supply values are anomalous, a 28-day supply is assumed.			Applied to T_MEDICAL claims where DAYS_SUPPLY is NULL. For T_RX pharmacy claims, DAYS_SUPPLY is typically populated but when missing/anomalous, 28 days is substituted. The assumed supply extends MAP_END_DT = FST_DT (or FILL_DT) + days_supply - 1.		JM 06Apr/2026
	LOT1_BASE_LENGTH	LOT1 base period duration in days		The protocol does not define LOT1_BASE_LENGTH as a named variable. LOT1 base period duration in days. Computed as LOT1_BASE_END_DT - LOT1_START_DT + 1. LOT start date is defined per Section 5.1.1 of the protocol. LOT end date is the priority-based end date defined in the LOT1_BASE_END_DT row.			Duration computed from dates already derived from T_MEDICAL/ RX (LOT1_START_DT from first MAP_START_DT) and LOT1_BASE_END_DT (priority-based end date).		JM 06Apr/2026
	PERMISSIBLE_SUBS_EFFECT	Permissible substitution rules		Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered			Substitutions tracked via CL_MM4_ROLLUP (Tab 40) which maps medication abbreviations. The permissible_subst reference table pairs		

Substitutions tracked via CL_MMA_ROLLUP (Tab 40) which maps medication abbreviations. The permissible_subs reference table pairs original_med with substitute_med. When a new medication claim appears in T_MEDICAL_TX, it is checked against the permissible_subs table to determine if it is a substitution of a tracked medication. Substitutions are tracked via CL_MMA_ROLLUP (Tab 40) which maps medication abbreviations. The permissible_subs reference table pairs original_med with substitute_med. When a new medication claim appears in T_MEDICAL_TX, it is checked against the permissible_subs table to determine if it is a substitution of a tracked medication.

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PERMISSIBLE_SUBS_EFFECT		Permissible substitution rules affecting LOT1 end					Substitutions tracked via CL_MMA_ROLLUP (Tab 40) which maps medication abbreviations. The permissible_subs reference table pairs original_med with substitute_med. When a new medication claim appears in T_MEDICAL_TX, it is checked against the permissible_subs table to determine if it is a substitution of a tracked medication.	
FU_PD		LOT1_TX_ENDDATE_REASON (AUTO/ALLO/CART)					SCT type determined from T_MEDICAL procedure codes matched against CL_SCT_CODELIST (Tab 44) which classifies procedures as AUTO, ALLO, or CART. Priority is earliest LOT-ending date wins.	1M09/205
FU_PD		LOT1_TX_AUTO_DT_1/ LOT1_TX_AUTO_DT_2	Dates of first and second valid autologous HSCT				AUTO HSCT dates derived from T_MEDICAL procedure codes (PROC_CD, BILL_PROC_CD) matched against CL_SCT_CODELIST where SCT_TYPE= AUTO. FST_DT from T_MEDICAL is the raw claim date. 14-day windowing groups claims; 60-day gap validation separates distinct events.	1M09/205
FU_PD		LOT1_BASE_DISCON_DT	LOT1 base period discontinuation date				max(MAP_END_DT) derived from T_MEDICAL (FST_DT + medical_day_supply - 1) and T_RX (FILL_DT + DAYS_SUP - 1) for induction medications matched via CL_MMA_CODELIST. Gap validated against OBS_END_DT from T_MEMBER_CONT_ENROLLMENT/ DOD.	1M09/205
FU_PD		LOT1_BASE_IST_ADD_MED_LOT1_BASE_IST_ADD_MED	Date and name of first non-induction medication added				First non-induction medication identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MMA_CODELIST (Tab 41) where the medication is NOT in base_meds (induction + permissible subs). MAP_START_DT from MAP_algorithm. Applied to MAP_START_DT within the window are classified as induction via CL_MMA_CODELIST/CL_MMA_ROLLUP matching.	1M09/205
FU_PD		INDUCTION_WINDOW_DAYS	Induction regimen identification window (days)				ALLO SCT identified from T_MEDICAL procedure codes matched against CL_SCT_CODELIST where SCT_TYPE=ALLO. Date from FST_DT in T_MEDICAL.	1M09/205
FU_PD		ALLO_ALWAYS_ENDS_LOT	Allogeneic SCT always ends current LOT (protocol rule)					1M09/205
FU_PD		CART_45D_CONSOLIDATION	CAR-T 45-day supportive agent consolidation rule				Would require checking T_MEDICAL_TX claims within 45 days of FIRST_CART_DT and reclassifying them from LOT1 to the CAR-T LOT. Affects MAP_END_DT for induction meds and MAP_START_DT for potential new meds.	1M09/205
FU_PD		NAMING_DISCREPANCY_SCT_VS_HSCT	Variable naming: spec uses HSCT prefix vs code uses SCT prefix				N/A - naming convention only.	1M09/205
FU_PD							Set contains_mtx_reg = 1 when BOTH of the following are true for the LOT1 induction regimen: (a) The induction regimen contains at least one valid mono-maintenance agent OR a valid dual-maintenance combination (see lists below). AND (b) At least one additional agent is present outside that qualifying maintenance subset, acting as an anchor. Otherwise contains_mtx_reg = 0 Valid mono-maintenance agents: lenalidomide, bortezomib, daratumumab, ixazomib, thalidomide	OK18/0205

